Midterm Revision Worksheet — Biology Grade 10

Lesson 7.4 Structures and Organelles · Lesson 8.3 Cellular Respiration · 31 questions

Name: _____ **Class:** _____ **Date:** _____

Section A · Multiple Choice

Questions 1–17

Circle the letter of the best answer.

1. Which organelle is called the "powerhouse of the cell"?

A ribosome **B** vesicle
C nucleus **D** mitochondria

2. Which organelle is large in plant cells but small or absent in animal cells?

A vacuole **B** chloroplast
C centriole **D** nucleolus

3. What is the function of the smooth ER?

A Proteins are synthesized and produced.
B A variety of complex carbohydrates and lipids are synthesized.
C It surrounds the nucleus with a double membrane.
D Carbohydrates, lipids, and other substances are consumed.

4. The stroma is the fluid-filled space where the _____ take place.

A chlorophyll intake **B** bond breaking reactions
C light-dependent reactions **D** light-independent reactions

5. Which organelle is present in a paramecium protozoan but absent in the cells of a strawberry plant?

A microtubules **B** nucleus
C cytoskeleton **D** cilia

6. Which structure has this role in the cell?
"Modifies, sorts, and packages proteins"

A ribosome **B** vacuole
C Golgi apparatus **D** endoplasmic reticulum

7. Which structure has this role in the cell?
"Contains substances that digest worn-out organelles and food"

A mitochondrion **B** vacuole
C lysosome **D** Golgi apparatus

8. Which organelle produces protein for a cell?

A ribosome **B** nucleus
C endoplasmic reticulum **D** Golgi apparatus

9. Which structure matches this function?
"Organelles that are made of microtubules that function during cell division"

A cytoskeleton **B** centrioles
C cilia **D** microtubules

10. Which describes the process of cellular respiration?

A anabolic pathway that breaks molecules into energy
B catabolic pathway that breaks molecules into energy
C catabolic pathway that builds molecules of stored energy
D anabolic pathway that builds molecules of stored energy

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11. Why is there a net yield of two, not four, ATP molecules in glycolysis?

A 2 molecules of ATP are used to begin glycolysis.
B 2 molecules of ATP are produced from ATP synthase.
C 2 molecules of ATP are produced during glycolysis.
D 2 molecules of ATP are used to complete glycolysis.

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12. Under ideal conditions, how many ATP molecules are created from cellular respiration in eukaryotic cells?

A 32
B 34
C 36
D 38

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13. Why is the electron transport chain an aerobic process?

A The electron transport chain produces carbon dioxide in its steps.
B The electron transport chain utilizes oxygen in its steps.
C The electron transport chain does not utilize oxygen in its steps.
D The electron transport chain does not produce carbon dioxide in its steps.

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14. Why is glycolysis an anaerobic process?

A Glycolysis produces carbon dioxide in its steps.
B Glycolysis does not produce carbon dioxide in its steps.
C Glycolysis utilizes oxygen in its steps.
D Glycolysis does not utilize oxygen in its steps.

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15. Which describes the process of fermentation?

A aerobic process that manufactures pyruvate
B anaerobic process that manufactures pyruvate
C anaerobic process that restores NAD^+ supply
D aerobic process that restores NAD^+ supply

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16. Which process does not occur in the mitochondria?

A glycolysis
B electron transport chain
C Krebs cycle
D preparatory reaction

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17. What is the importance of NADH and $FADH_2$?

A These molecules are useless by-products.
B These molecules are recycled for use in glycolysis.
C These molecules are used in the electron transport chain.
D These molecules are used as products in photosynthesis.

Section B · Modified True or False

Questions 18–21

Circle T or F. If the statement is false, write the word(s) that should replace the underlined word to make it true.

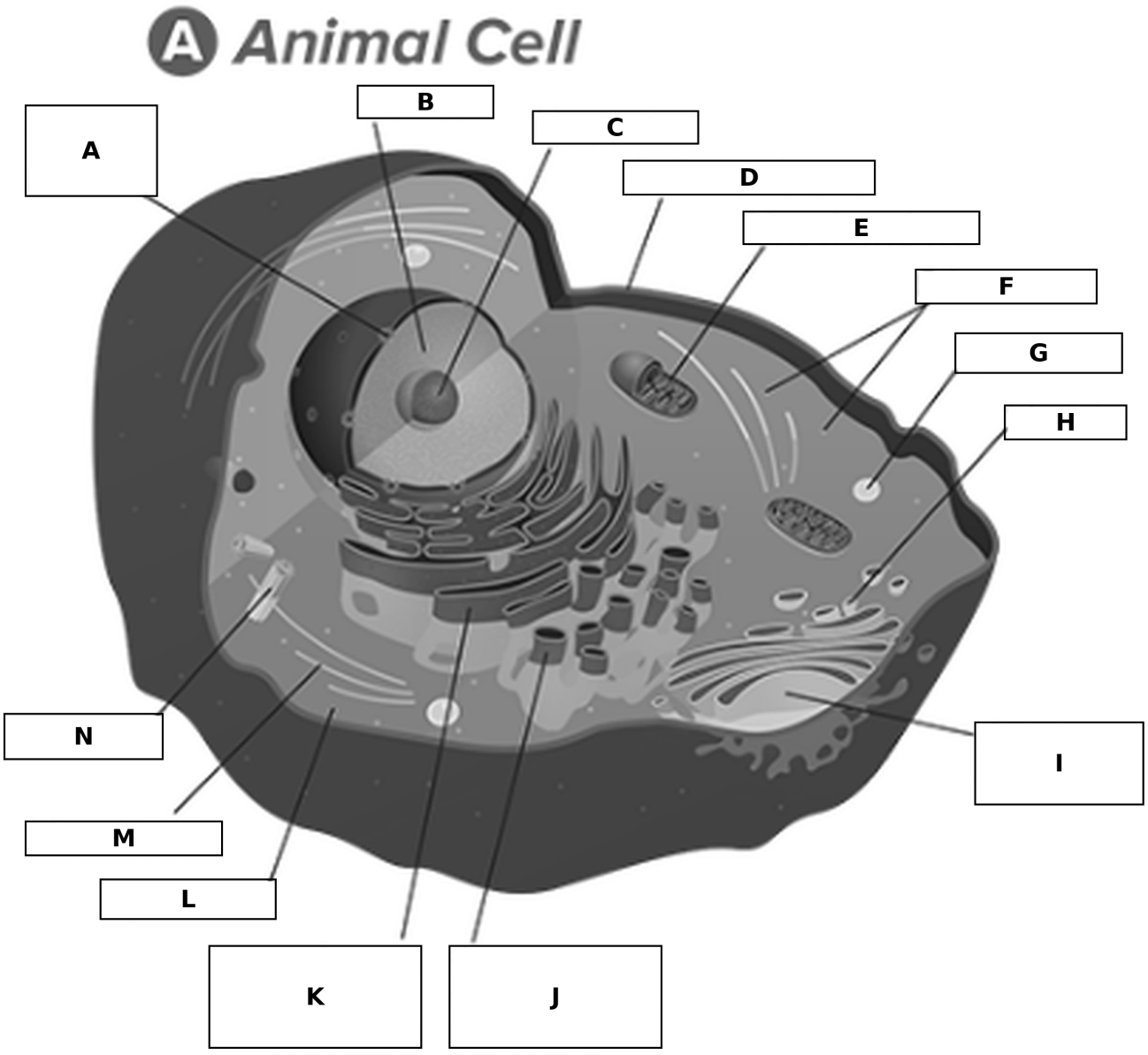
		Circle	Correction (if false)
18.	The chloroplasts harness light energy and convert it into chemical energy.	**T F**	_____
19.	The central structure that defines the cell is called the cytoplasm.	**T F**	_____
20.	Energy needed to maintain body temperature despite ongoing energy transfer to the environment is released.	**T F**	_____
21.	The final step of aerobic respiration when ATP is created is called glycolysis.	**T F**	_____

Section C · Label the Figures

Questions 22–25

Write the name of every lettered structure on the matching line.

22. Label the animal cell.



A _____ B _____ C _____

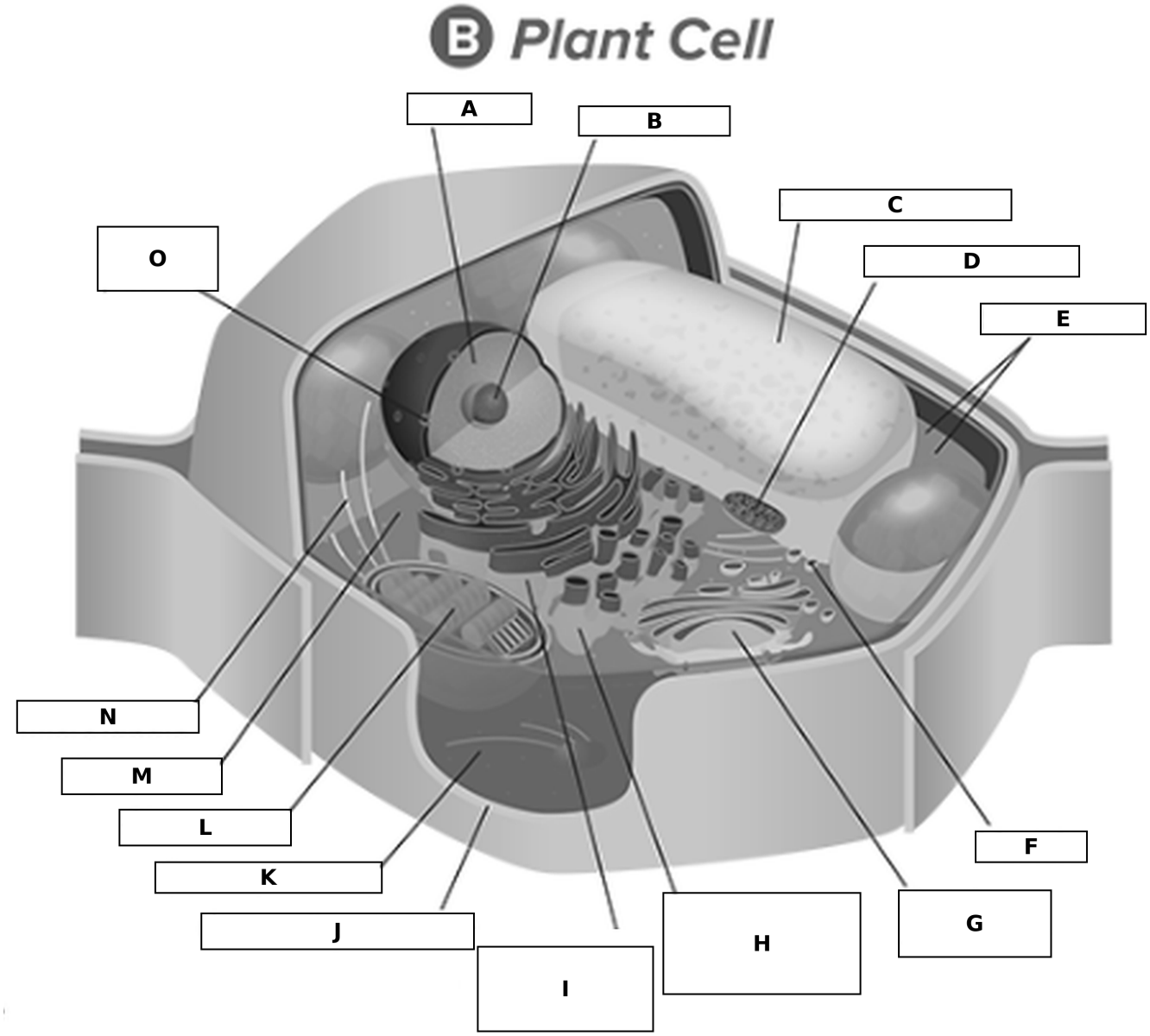
D _____ E _____ F _____

G _____ H _____ I _____

J _____ K _____ L _____

M _____ N _____

23. Label the plant cell.



A _____ B _____ C _____

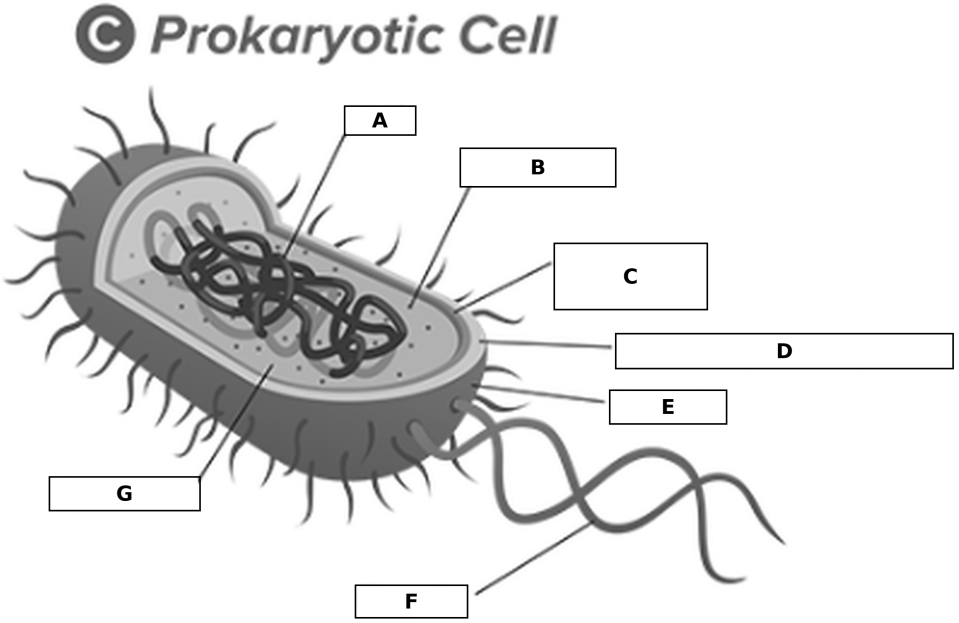
D _____ E _____ F _____

G _____ H _____ I _____

J _____ K _____ L _____

M _____ N _____ O _____

24. Label the bacterial (prokaryotic) cell.

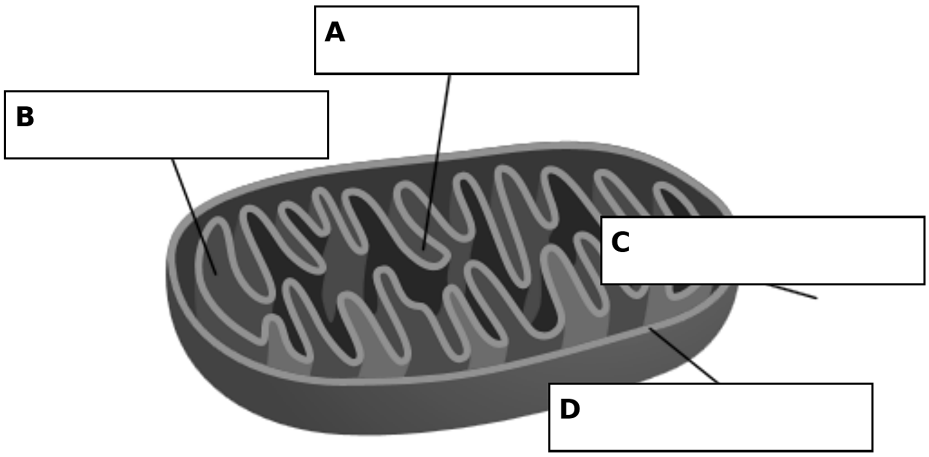


A _____ **B** _____ **C** _____

D _____ **E** _____ **F** _____

G _____

25. Write the structures in the correct location in the mitochondria.



Word bank: Cristae · Matrix · Outer membrane · Inner membrane

Section D · Complete

Questions 26–31

Write the missing word(s) on the line.

26. Complete the chemical equation for cellular respiration. Use the chemical formulas (e.g. H_2O).

_____ + 6 _____ → 6 _____ + 6 H_2O + _____

27. Identify how many carbons from one glucose molecule enter one round of the Krebs cycle.

Answer: _____

28. **Skeletal muscles produce _____ when the body cannot supply enough oxygen, such as during periods of strenuous exercise.**

29. **Alcohol fermentation occurs in _____ and some bacteria.**

30. **The Krebs cycle also is referred to as the _____ cycle.**

31. **The H^+ ions are pumped into the _____ through the proteins in the inner mitochondrial membrane.**

Answer Key

Questions 1–31

Answers only.

Section A

Q	Answer	Q	Answer	Q	Answer
1	D mitochondria	7	C lysosome	13	B utilizes oxygen
2	A vacuole	8	A ribosome	14	D does not utilize oxygen
3	B carbohydrates and lipids synthesized	9	B centrioles	15	C anaerobic … restores NAD^+ supply
4	D light-independent reactions	10	B catabolic … into energy	16	A glycolysis
5	D cilia	11	A 2 ATP used to begin glycolysis	17	C used in the electron transport chain
6	C Golgi apparatus	12	C 36		

Section B

Q	T / F	Correction
18	T	—
19	F	nucleus
20	T	—
21	F	electron transport (chain)

Section C

22. Animal cell: A nuclear pore · B nucleus · C nucleolus · D cell membrane · E mitochondrion · F ribosomes · G lysosome · H vesicle · I Golgi apparatus · J smooth endoplasmic reticulum · K rough endoplasmic reticulum · L cytoplasm · M microtubule · N centriole

23. Plant cell: A nucleus · B nucleolus · C central vacuole · D mitochondrion · E ribosomes · F vesicle · G Golgi apparatus · H smooth endoplasmic reticulum · I rough endoplasmic reticulum · J cell wall (cellulose) · K cell membrane · L chloroplast · M cytoplasm · N microtubule · O nuclear pore

24. Prokaryotic cell: A DNA · B ribosomes · C plasma membrane · D cell wall (peptidoglycan) · E capsule · F flagella · G cytoplasm

25. Mitochondria: A cristae · B matrix · C outer membrane · D inner membrane

Section D

26. $C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O$ + energy

27. 2 carbons (as acetyl CoA) **28.** lactic acid **29.** yeast **30.** citric acid (also: tricarboxylic acid, TCA) **31.** intermembrane space